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Debre Berhan University

Practical Education for a Better Success!
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DEPARTMENT OF MATHEMATICS

Numerical Analysis

Practice Midterm Examination

Math2081 and Math3051 | Chapters 1–3

Time allowed: 2 hours **Total:** 30 marks (30%)

Name: _____ **Student ID:** _____

Section: _____ **Date:** _____

Instructions

- Answer all 25 questions. A scientific calculator is permitted.
- For Parts I and III, write your answer in the space provided. Complete each blank in Part II.
- In Part IV, show formulas, substitutions and intermediate steps. Use additional paper if needed.
- Retain at least six decimal places during calculations. Report final numerical answers to four decimal places unless significant figures or exact values are requested. Use exact fractions or scientific notation when four decimal places would hide a small error.

Part I True or False **2.5 marks**

Write **True** or **False**. Each question carries **0.5 mark**.

1. _____ Relative error with the exact value in the denominator is undefined when the exact value is zero.

2. _____ Newton–Raphson converges to a root for every initial approximation.

3. _____ Strict row diagonal dominance is a sufficient condition for convergence of both Jacobi and Gauss–Seidel iterations.

4. _____ A small residual necessarily implies a small solution error, regardless of the conditioning of the coefficient matrix.

5. _____ The standard regula falsi method can retain the same endpoint for many consecutive iterations.

Part II Fill in the Blanks **2.5 marks**

Complete each statement or formula. Each question carries **0.5 mark**.

6. Newton–Raphson iteration for $f(x) = 0$ is $x_{n+1} =$ _____.

7. The secant iteration formula is $x_{n+1} =$ _____.

8. For $Ax = b$, the residual corresponding to an approximate solution $\tilde{x}$ is $r =$ _____.

9. For $x^{(k+1)} = Tx^{(k)} + c$, convergence to the unique fixed point for every initial vector occurs if and only if $\rho(T)$ _____, where $\rho(T)$ is the spectral radius.

10. In Newton's method for $F(x) = 0$, the correction $s^{(k)}$ satisfies

$$J_F\left(x^{(k)}\right) s^{(k)} = _____, \qquad x^{(k+1)} = x^{(k)} + s^{(k)}.$$

Part III Multiple Choice **10 marks**

Choose **one** correct answer for each question and write its letter in the blank. Each question carries **1 mark**.

11. If p is an exact value and $\tilde{p}$ is its approximation, the absolute error is: _____

A. $p - \tilde{p}$ B. $|p - \tilde{p}|$

C. $\left|\dfrac{p - \tilde{p}}{p}\right|$ D. $\left|\dfrac{\tilde{p}}{p}\right|$

12. The number 7.28651, rounded to four significant figures, is: _____

A. 7.286 B. 7.287

C. 7.290 D. 7.285

13. Truncation error occurs when: _____

A. An infinite mathematical process is replaced by a finite approximation.
B. A number is entered incorrectly into a calculator.
C. An exact number is stored using a limited number of digits.
D. Two rows of a matrix are interchanged.

14. Which conditions guarantee at least one root of $f(x) = 0$ in (a, b)? _____

A. $f(a) = f(b)$.
B. f is differentiable and $f'(a) > 0$.
C. f is continuous on $[a, b]$ and $f(a)f(b) < 0$.
D. $f(a)f(b) > 0$.

15. Which method always halves the current bracketing interval at each iteration? _____

A. Newton–Raphson B. Secant

C. Regula falsi D. Bisection

16. The regula falsi approximation using endpoints a and b is: _____

A. $\dfrac{a+b}{2}$ B. $a - \dfrac{f(a)}{f'(a)}$

C. $\dfrac{af(b) - bf(a)}{f(b) - f(a)}$ D. $\dfrac{f(a) + f(b)}{2}$

17. Suppose g is continuously differentiable near a fixed point α. Which condition guarantees local convergence of $x_{n+1} = g(x_n)$ for sufficiently close initial guesses? _____

A. $|g'(\alpha)| < 1$ B. $|g'(\alpha)| > 1$

C. $g(\alpha) = 0$ D. $g'(\alpha) = 2$

18. Which method normally starts with two initial approximations and does not require an explicit derivative? _____

A. Newton–Raphson B. Bisection with only one endpoint

C. Gaussian elimination D. Secant

19. For the system $4x + y = 1$, $2x + 3y = 2$, one Jacobi iteration from $(0, 0)$ gives: _____

A. $(1/4, 1/2)$ B. $(1/4, 2/3)$

C. $(1/12, 1/2)$ D. $(0, 0)$

20. The main distinction between Gauss–Seidel and Jacobi iteration is that Gauss–Seidel: _____

A. Requires the matrix inverse at each iteration.
B. Works only for diagonal matrices.
C. Uses newly computed components immediately within the current iteration.
D. Always converges for every nonsingular matrix.

Part IV Workout Problems **15 marks**

Show your working clearly. Method marks are awarded for correct intermediate steps.

21. Numerical errors and finite precision **[2 marks]**

(a) The exact value is $p = \frac{1}{3}$, and its approximation is $\tilde{p} = 0.333$. Calculate the absolute error, the relative error and the percentage relative error. Give exact fractions or at least three significant figures for the errors. **[1.5 marks]**

(b) For 0.00456789, find its chopped value and its rounded value to **four significant figures**. **[0.5 marks]**

22. Bisection method **[3 marks]**

Consider $f(x) = x^3 - x - 2$ on the interval $[1, 2]$.

(a) Show that the interval contains at least one root. **[0.5 marks]**

(b) Perform **three bisection iterations**. Record the endpoints before each iteration, midpoint, midpoint function value and retained interval in the table below. **[1.5 marks]**

(c) State the third midpoint approximation and a guaranteed upper bound for its absolute error. **[0.5 marks]**

(d) Let c_n denote the midpoint evaluated at iteration n, with c_1 the first midpoint of $[1, 2]$. Find the smallest n for which the standard bisection error bound guarantees $|c_n - \alpha| \leq 10^{-3}$, where α is the root. **[0.5 marks]**

Iteration	a	b	c_n	$f(c_n)$	**Retained interval**
1					
2					
3					

23. Newton–Raphson method **[3 marks]**

Use Newton–Raphson to approximate the positive root of $f(x) = x^2 - 2$.

(a) Derive the iteration formula. **[1 marks]**

(b) Starting with $x_0 = 1.5$, calculate x_1 and x_2. **[1.5 marks]**

(c) Explain why the standard Newton update cannot be evaluated at $x_0 = 0$. **[0.5 marks]**

Part IV Workout Problems Continued

24. Jacobi and Gauss–Seidel methods [5 marks]

Consider

$$\begin{aligned} 4x + y &= 1, \\ 2x + 3y &= 2, \end{aligned} \qquad \text{with } (x^{(0)}, y^{(0)}) = (0, 0).$$

(a) Write the Jacobi iteration formulas. **[1 marks]**

(b) Perform two Jacobi iterations. **[1 marks]**

(c) Write the Gauss–Seidel iteration formulas, updating x before y, and perform two iterations. **[1.5 marks]**

(d) Verify that the coefficient matrix is strictly row diagonally dominant. State the implication for convergence. **[0.5 marks]**

(e) For each method's second iterate, calculate $\|r^{(2)}\|_\infty$, where

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad u^{(2)} = \begin{pmatrix} x^{(2)} \\ y^{(2)} \end{pmatrix}, \qquad r^{(2)} = b - Au^{(2)}.$$

Which method has the smaller residual norm after these two iterations? **[1 marks]**

	Jacobi		**Gauss–Seidel**	
Iteration	$x^{(k)}$	$y^{(k)}$	$x^{(k)}$	$y^{(k)}$
0	0	0	0	0
1				
2				

25. Newton iteration for a nonlinear system [2 marks]

Consider

$$F(x, y) = \begin{pmatrix} x^2 + y^2 - 2 \\ x - y \end{pmatrix}, \qquad u^{(0)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

(a) Find the Jacobian $J_F(x, y)$ and evaluate both $J_F(u^{(0)})$ and $F(u^{(0)})$. **[1 marks]**

(b) Solve $J_F(u^{(0)})s^{(0)} = -F(u^{(0)})$ and calculate $u^{(1)} = u^{(0)} + s^{(0)}$. Perform only one Newton iteration. **[1 marks]**

End of practice examination